

# HEESOO JUNG

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## PROFILE

Ph.D. student in machine learning with 12 peer-reviewed publications (6 first/co-first author) at KDD, WWW, WSDM, and ACL. Research spans large language models (LLM personalization and multi-agent communication with Samsung Research), reinforcement learning for GNNs, and time-series sensing for healthcare. Seeking a research-focused ML internship.

## EDUCATION

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|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Ph.D., Computer Science and Engineering</b><br><i>Sungkyunkwan University (SKKU) · Advisor: Prof. Hogun Park · GPA 4.19/4.5</i>         | Sep 2023 – Present<br>Suwon, KR  |
| <b>M.S., Artificial Intelligence</b><br><i>Sungkyunkwan University (SKKU) · GPA 4.44/4.5</i>                                               | Sep 2021 – Aug 2023<br>Suwon, KR |
| <b>B.S., Computer Science and Mechanical Engineering (Double Major)</b><br><i>Sungkyunkwan University (SKKU) · Cum Laude, GPA 4.24/4.5</i> | Mar 2017 – Aug 2021<br>Suwon, KR |

## RESEARCH EXPERIENCE

|                                                                                                                     |                                 |
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| <b>Samsung Electronics (Samsung Research)</b><br><i>Industry-Academia Research Collaboration</i>                    | 2024 – Present                  |
| • <b>LLM Personalization:</b> Updated personalized knowledge in large language models via activation steering.      | 2024 – 2025                     |
| • <b>User Log Mining:</b> Extracted behavioral patterns from historical user logs.                                  | 2025 – 2026                     |
| • <b>Multi-Agent LLMs:</b> Designing agent-to-agent communication to clarify ambiguous user requests.               | 2026 – Present                  |
| <b>Kangbuk Samsung Hospital</b><br><i>Clinical Research Collaboration</i>                                           | Mar 2024 – Present              |
| • <b>Dysphagia Detection:</b> Built a deep learning model to detect dysphagia non-invasively from sEMG signals.     |                                 |
| <b>Graduate Researcher, LearnData Lab</b><br><i>Sungkyunkwan University (SKKU)</i>                                  | Sep 2021 – Present<br>Suwon, KR |
| • Authored 12 peer-reviewed papers (6 as first/co-first author) spanning GNN explainability and KG reasoning.       |                                 |
| • Advanced graph self-supervised learning via masked auto-encoders (WSDM 2025) and smoothness balancing (WWW 2025). |                                 |
| • Received Innovation SKKU Fellowship Award (2025, 2026)                                                            |                                 |

## SELECTED PUBLICATIONS

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|--------------------------------------------------------------------------------------------------------------------|----------------------------|
| • Out-of-Distribution Robust Explainer for Graph Neural Networks                                                   | KDD 2026                   |
| • Harnessing Spatial Dependency for Domain Generalization in Multivariate Time-Series Sensor Data                  | ESWA 2026                  |
| • Harnessing Influence Function in Explaining Graph Neural Networks                                                | KDD 2025                   |
| • Balancing Graph Embedding Smoothness in Self-Supervised Learning via Information-Theoretic Decomposition         | WWW 2025                   |
| • CIMAGE: Exploiting the Conditional Independence in Masked Graph Auto-Encoders                                    | WSDM 2025                  |
| • Complex Reasoning in Knowledge Graph Question Answering Through Query Graph Approximation                        | ACL Findings 2025          |
| • Enhancing Inductive Numerical Reasoning in Knowledge Graphs with Relation-Aware Relative Numeric Encoding        | PAKDD 2025                 |
| • Improving Multi-hop Logical Reasoning in Knowledge Graphs with Context-Aware Query Representation Learning       | ACL Findings 2024          |
| • Dual Policy Learning for Aggregation Optimization in GNN-based Recommender Systems                               | WWW 2023                   |
| • Exploiting Relation-aware Attribute Representation Learning in Knowledge Graph Embedding for Numerical Reasoning | KDD 2023                   |
| • Stretchable Array sEMG Sensor with Graph Neural Network for Static and Dynamic Gesture Recognition               | npj Flex. Electron. 2023   |
| • sEMG-based Gesture Recognition with Deep Neural Networks                                                         | IEEE BigComp Workshop 2022 |

## PATENT APPLICATIONS

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| • Determining the Number of GNN Layers Using a Reinforcement Learning Model · US 18/090,685 · KR 10-2022-0129721 | 2022 |
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## ACADEMIC SERVICE

**Session Chair:** KDD 2025

**Program Committee:** The Web Conference (WWW) 2025, Short Paper Track

**Reviewer:** NeurIPS 2025; ICLR 2026; KDD 2024, 2025 (Excellent Reviewer Award), 2026; WWW 2025, 2026

## SKILLS

**Programming & Tools:** Python, PyTorch, TensorFlow, PyTorch Geometric (PyG), DGL, Git

**Research Areas:** Graph Neural Networks, Recommender Systems, Knowledge Graphs, Self-Supervised Learning, Explainable AI

**Languages:** English (professional proficiency)